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Data Semantics: Course Projects

Guidelines

25-26 Edition

data * AI

AI * data

Remember: Assignments!

- Assignments are necessary and part of the evaluation
  - +/- 3 points if all done
- Assignments are discussed during the same exam where you present your project
  - You simply come with assignments
  - You will be asked to show the assignments and discuss your solutions and - if needed - answer questions on the topics related to the assignments
- Supporting documents
  - If you want, you can write a document to track what you did and your discussion, but it is not required
  - If you have such a document, remember to check it before the assignment and not to forget its content

Course Projects: Overview

- Groups: 1-4 people (2-3 are the magic numbers)
- Different Projects on Different Topics
  - Understand them / ask questions if needed
  - Choose one from the pool and/or combine them
  - Only if you are ultra confident on your ideas: propose
- Remind the goal of the project
  - Learn methodologies and tasks with more practical (but not sloppy) work
  - Understanding is more important than performance
    - Read related work / more on this in next slides
    - Evaluate your work - as allowed by expected project effort (sample-based can be ok)
    - Reflect on pros / limitations of the proposed approach

How to Define / Track Projects #1

- 1 meeting for discussing projects and Q&A
  - June 9th – **9.15** - 10.30 – A room in U14/U24 TBD
  - In presence – could do hybrid upon requests
- Process:
  - Specify your proposal in the forum "Data Semantics 25-26 - Course Projects"
  - Attend the meetings. You can prepare a short presentation to discuss the idea and receive feedback
  - Post your proposal (after Q&A, if needed) as a thread in the dedicated forum, including title, abstract, papers (which can be refined via interactions)
  - Follow the interactions (take a look at interactions for similar projects) and complete the definition - until: best before June 20th.
  - Track your final proposal in the spreadsheets "Data Semantics 25-26 - Course Projects" summary and project-specific sheets
  - Delivery – until: whenever you plan to give the exam

How to Define / Track Projects #2

- Shared spreadsheet to track projects
  - Data Semantics 24-25 - Course Projects Spreadsheet
  - Define your proposal in the Summary sheet first (examples are provided)
  - While you progress in the definition of your project via the forum, fill in the details about your project by copying the “Template Project Details” sheet into a new sheet with your project name. Also for this, an example is provided.
- “Course Projects” on the website
  - Data Semantics 24-25 - Course Projects Forum
  - All interactions will be managed through the website; so that all students can learn from Q&A on your project
  - In case emails are really needed, please write to all the involved teachers (Matteo Palmonari, Riccardo Pozzi)
- Delivery
  - Send an email to all teacherS with: final list of students + title + abstract: 1 week before the exam
  - Upload the slides anytime before the exam in the project folder:
  - https://drive.google.com/drive/folders/1At1QcSfpLZliG0G6wC9WQ3DLhj1xolCD?usp=sharing

Some examples from the previous year

- You can check some examples of presentations here:
  - 2024-2025
  - 2023-2024
  - 2022-2023 (only a subset of the presentations)
  - See also examples from previous years linked in the website
- Scores are not reported

All Projects: Reference Papers!

- Don’t reinvent the wheel!
  - If possible: reuse existing work/models/code
    - But…. no black box approach: understand how it works, its benefits, its limitation… if you have time left, try to improve it!
  - Otherwise: be inspired!
    - Something related to your research problem must exist. Read that and draw inspiration / methodologies / etc.
- Thus: every project must come with a selection of papers that students must read
  - 3-6 depending on the group size / problem (excl. course lessons)
  - Papers addressing …
    - the very same problem (reuse approach / evaluation etc,)
    - similar / close problem (reuse methodology)
  - Attention to “why questions”
    - why to / not to reuse? why different? …

1 - Declarative Semantics: Ontologies

- Design an ontology for a specific use case
  - From requirements to design in Protégé
    - Summarize requirements in the target application
    - Review existing ontologies covering similar problems / domains – possibly extend them
    - Think about the data to be modeled: manually add examples for each property/class in Protégé and use them to explain / test modeling
  - Explain and justify choices
    - Use a reasoner to check and discuss inferences
    - Discuss specific modeling choices also using inferences
- To read
  - Ontology design methodologies (euBusinessGraph ontology technical document + some theses)
  - Papers describing ontologies / semantic modeling in near domains.

2 - Declarative Semantics: Build a KG

- Design a KG for a specific use case from legacy data
  - From (short) requirements to full prototype implementation
    - Summarize requirements in the target application
    - Find data to transform into RDF
      - Data from scraping / tables/RDBs etc.
    - Model the ontology (lighter modeling than in project of type 1)
    - Transform the data
  - Test usage
    - Queries / exploration (?)
- To read / practice with
  - Technology to put up a server to host the KG (e.g., Virtuoso)
  - Technology to map existing data to RDF (e.g., D2R from RDBs)
  - Papers describing the above solutions

3 – Matching

- Matching pipeline to find links
  - Describe the task from a technical point of view (semantics of links, datasets, etc)
  - Select dataset
    - [preferred] UNIMIB-provided datasets: google geotarget vs. geonames as in assignment but done more seriously with full-fledged evaluation. Complete links across X EU countries (X based on the group size). Store links in RDF.
    - Datasets used in state-of-the-art work → understand / performance
    - Novel datasets → apply / evaluate at small scale
  - Use / improve existing technologies
    - You can practice with neural approaches, e.g., DeepMatcher (or Ditto) and compare results
  - Test and evaluate
    - Quantitative evaluation (full-fledged vs small-scale depending on the data)
    - Qualitative evaluation (discuss errors → understand limitations)
- To read: domain-specific, ML-based, neural, etc.

4 – Information Extraction from Text

- Pipeline to extract information from texts or other unstructured/semistructured data sources
  - Describe the task from a technical point of view (what to extract, usage of extracted information, etc.)
    - NER / NEL
    - Relation extraction (we suggest focused and pattern-based extraction)
  - Select dataset
    - Datasets used in state-of-the-art work → understand / performance
    - Novel datasets → apply / evaluate at small scale
  - Use existing technologies
    - NER / NEL APIs // BERT-based baselines
    - Information extraction suites (e.g., Expert.ai Suite)
    - Discuss limitations and – when possible – try improvement
  - Test and evaluate
    - Quantitative evaluation (full-fledged vs small-scale depending on the data)
    - Qualitative evaluation (discuss errors → understand limitations)
- To read: related work (deeper than in our lessons)

5 – Corpus-based Analyses with Distributional Semantics

- Topics: bias, temporal drifts, etc.
- Techniques
  - We suggest selecting techniques from Gallegos & al. 2024 – (see LLM2 lesson)
  - You can also use techniques based on word embedding - presented more in in previous editions
    - Check the seminar: DS2223 - 4.4 - Some Applications to Computational Social Sciences - Evolution and Biases
    - CADE (+/- entities) WEAT / SWEAT (+/- entities)
- Corpora
  - Better used recommended corpora; you'll need to sign a document to commit not to share data

6 – ChatGPT or smaller models

- ChatGPTx … yes… it can fit under specific conditions, such as:
  - In depth study of novel topic (e.g., retrieval augmentation, toolformers, prompt learning, etc.)
  - Evaluate some task as performed by GPTX vs comparison… but in a systematic way
- Other smaller / open generative models…
  - LLAMA, QWEN, DeepSeek, Gemma…
  - Alpaca
  - … etc.

7 – RAG with ChatGPT or smaller models

- Same as before but
  - Document-based RAG
  - Graph RAG (any type)
  - … remember the evaluation
  - … specific RAG topic, e.g., retrieval, indexing, rewriting etc

8 – Proposed by Students

- Are you really sure you want to propose your own project? Think about it twice ☺
- Write your idea well and convince it makes sense in relation to recommended projects
  - Find data and profile their availability / size
  - Make sure that the size of the data you have match the requirements of the specific techniques (e.g., be careful with tweets for W2V or CADE)
  - Fine related work / papers!
- It makes sense if you are driven by personal passion / previous work (of course, extended in this project

DRILL-DOWN: DISTRIBUTIONAL SEMANTICS PROJECTS

Sketch of Proposed Projects

- Text analysis with novel measures based on LLMs (Gallegos & al. 2024), or word2vec-based (CADE, WEAT, SWEAT)
  - Select corpora
  - Define research questions
  - Check that corpora / techniques can support the target research question
    - Enough data?
    - Control for word frequency and sample size (comparable)
    - Be inspired by similar analyses that are published (also a requirement: see assigned papers)

Proposed Project: Example Sketch

- Corpus: Cinema
- Analyses / research questions
  - Diachronic evolution of target words:
    - Top changers / top stables
    - Semantic change classification with threshold
      - Compare distant slices: what has changed? Review / discuss manually
    - Semantic evolution
      - Track evolution along time: what has changed most?
    - Drill-down on specific words/entities
      - Socio-economic: e.g., rich / poor; political: e.g., democratic / republican; gender (follow best practices, e.g., Caliksan&al.)
  - Similar for other dimension (e.g., movies by genre)

Corpora

From **corpusdata.org**
For more information check:

https://www.corpusdata.org/corpora.asp

Most of them new for us, you need a bit of explorative analysis.

Restriction on usage:
- Compile a form
- Commit not to share anywhere on the web or with other people
- Usage limited to the scope of the project

Corpus	Texts (95% available in full-text data)	Focus / strengths
iWeb: The Intelligent Web Corpus (More info)	14 billion words / 22 million web pages / ~100,000 websites	Size, size, and more size. Taken from ~100,000 of the most widely-used websites (for English) in the world. Probably the best for "web / tech" language
NOW: News on the Web (Two datasets; more info)	billion words / 0 texts. (Still growing every month; last update is for Apr 2021)	The most up-to-date corpus of English. Wide range of online newspapers and magazines (technology, entertainment, sports, politics, etc)
Coronavirus Corpus (Two datasets; more info)	million words / 0 texts. (Still growing every month; last update is for Apr 2021)	Designed to be the definitive record of the social, cultural, and economic impact of the coronavirus (COVID-19) in 2020 and beyond.
COCA: Corpus of Contemporary American English (More info)	1 billion words / 485,000 texts. US, 1990-2019	Best coverage of all types of genres (informal to formal): TV/Movies subtitles, blogs, web pages, spoken, fiction, magazines, newspaper, academic. The most widely-used corpus of English.
GloWbE: Global Web-based English	1.9 billion words / 1.8 million texts. 20 countries	About 60% blogs (very informal). Recent: 2013. Comparing varieties of English: American, British, Australian, etc. 100x as large as the next-largest corpus of English dialects.
~~Wikipedia Corpus~~	~~1.9 billion words / 4.4 million texts~~	~~Best corpus for specialized language for an almost~~ unlimited range of topics: science, entertainment, technology, history, sports, etc
COHA: Corpus of Historical American English	400 million words / 107,000 texts. US, 1810-2009	Historical change. 100x as large as next-largest historical corpus of English.
TV Corpus	325 million words / 75,000 episodes. US, UK, 4 other dialects, 1950-2018	Extremely informal language (more info). Can also be used to compare dialects and changes since the 1950s.
Movies Corpus	200 million words / 25,000 movies. US, UK, 4 other dialects, 1930-2018	Extremely informal language (more info). Can also be used to compare dialects and changes since the 1930s.

New York Times

- Used for TWEC experiments

Training Data	Authors	# Articles	Time Span	# Slices
Small Corpus	[Yao+,2018]	99,872	1990-2016	27
Large Corpus	[Sandhaus, 2008]	1,855,658	1987-2007	21

- Include analogies for temporal word embeddings

Test Data	Authors	# Analogies	Time Span	# Categories
Testset 1	[Yao+,2018]	11,028	1990-2016	25
Testset 2	[Szymanski,2017]	4,200	1987-2007	10

Semantic Change Detection Data

- International challenge (SemEval2020)

	T1		**T2**	
Language	**Corpus**	**Tokens**	**Corpus**	**Tokens**
English	1810-1860	6.5M	1960-2010	6.7M
German	1800-1899	70.2M	1946-1990	72.3M
Latin	-200-0	1.7M	0-2000	9.4M
Swedish	1790-1830	71.M	1895–1903	110M

- Italian challenge (Eval-Ita 2020)
  - Italian

Reddit Boards

- Short-term meaning shift data: Liverpool 11-13 vs Liverpool 17 (Reddit)

  (+ Generic 11-13)
  - M. Del Tredici, R. Fernández, and G. Boleda. Short-term meaning shift: A distributional exploration. In NAACL 2019 Volume 1 (Long and Short Papers).
  - Veronica Leorati MA thesis as a reference
  - Small data, several interesting phenomena
- Many more boards: check Nicoli's thesis for reference

Additional datasets / topics

- We do not know the following datasets well. Check size before using them.
  - Dataset of UK parliament debates
    - https://hansard.parliament.uk
  - Dataset of political parties manifests
    - https://manifesto-project.wzb.eu/ (multilanguage)